

Rezwana Razzaque Angana

Location: 3 Sunshine Avenue, Karori, Wellington 6012

Email: rezwana.angana@vuw.ac.nz

Mobile: +64 22 444 740811

DIGITAL PRESENCE

 github.com/anganarezw |  [linkedin.com/in/rezwana-angana](https://www.linkedin.com/in/rezwana-angana) |  [researchgate.net/profile/Rezwana-Angana](https://www.researchgate.net/profile/Rezwana-Angana)

PROFILE

I am a PhD candidate in Applied Mathematics at Victoria University of Wellington working on nonlinear dispersive wave phenomena. My research explores the mathematical structure of undular bores and weakly singular shock waves through Boussinesq-type and shallow water models. I combine rigorous analytical methods with computational simulations to investigate wave regularization, stability, and traveling wave dynamics.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, Mathematica

Numerical Methods: Finite difference methods, Finite Volume Methods, Finite Element Methods.

Areas: Partial differential equations, nonlinear waves, dynamical systems.

EDUCATION

Doctor of Philosophy (PhD) in Applied Mathematics

2023 – Present

Victoria University of Wellington (VUW), New Zealand

Thesis: *On the Mathematical Theory of Undular Bores*

Supervisor: Dr. Dimitrios Mitsotakis

Master of Science (MSc) in Mathematics

2015 – 2017

University of Dhaka, Bangladesh

WORK EXPERIENCE

Lecturer in Mathematics

2017

American International University-Bangladesh (AIUB), Dhaka, Bangladesh

Tutor and Marker in Mathematics

2024 – Present

Victoria University of Wellington (VUW), New Zealand.

Calculus I & II · Linear Algebra · Algebra · Discrete Mathematics · Mathematical Logic · Introduction to Mathematical Thinking · Engineering Mathematics (Logic & Statistics) · Statistics in Practice · Groups and Graphs

CONFERENCES AND WORKSHOPS

- **NZMASP 2025 (New Zealand Mathematics Association of Postgraduate Students)**, Auckland, New Zealand)
Oral presentation: *Heteroclinic Orbit Analysis of Traveling Wave Solutions in Dissipative Bona–Smith Systems*
- Poster Presentation: **International Conference on Advances in Computational Mathematics**, University of Dhaka, Bangladesh (May 27–28, 2017). Topic: An introduction to impulsive differential equations and their stability criteria using Lyapunov's direct method.
- Attended training workshop: **Capacity Building Training on Productivity Improvement Technique at Workshop Level**, jointly organized by Ministry of Commerce and Bangladesh Engineering Industry Owners Association (BEIOA).

SCHOLARSHIPS AND MEMBERSHIPS

- Wellington Doctoral Scholarship (November 2023-October 2026).
- Irene Pestov Memorial Scholarship 2026
- Lifetime Member, Bangladesh Mathematical Society.

PUBLICATIONS

- Rezwana Razzaque Angana, *Stability Criteria of Impulsive Differential Equations by Lyapunov's Second Method*, IOSR Journal of Mathematics, Vol. 16, Issue 2 (Mar–Apr 2020), pp. 01–03.
- Rezwana Razzaque Angana and Dimitrios Mitsotakis, *A high-order regularization of the non-linear shallow water equations with weakly singular shock waves and its approximation by finite volume methods*, Computational and Applied Mathematics, Springer, 2026. DOI:10.1007/s40314-026-03839-6.

INTERESTS

Cooking and developing new recipes; exploring diverse cuisines.

REFERENCES

- Available upon request.